

Website name : Herb and oil

System : Skincare Products E-Commerce System

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Bootcamp : Web development using Java and integration of artificial intelligence technologies.

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Introduction

The system is an online platform for selling skincare products. It allows users to access a variety of beauty products in one place with an easy and simple shopping experience.

System overview:

- Display skincare products
- Search for products by type or price
- Purchase product
- Order management
- Customer management
- Product management

Modules :

- Product module
- Customer module
- Order module

Product class :

Represents skincare products like creams, serums, and cleansers .

Variables :

- productName
- description
- price
- stock

Validation :

```
@NotBlank(message = "Product name is required")
```

```
@Size(max = 15 , message = "Product name must be less than 15 characters ")
```

```
private String productName ;
```

```
@NotBlank(message = "Product description is required")
```

```
@Size(min = 10, max = 200, message = "Description must be between 10 and 200  
characters")
```

```
private String description ;
```

```
@Positive(message = "Price must be greater than 0")
```

```
private double price ;
```

```
@PositiveOrZero(message = "Stock cannot be negative")
```

```
private int stock ;
```

Customer class :

Represents a customer who buys skincare products.

Variables :

- name
- password
- email
- phoneNumber

Validation :

@NotBlank(message = "Name can not be empty")

@Size(min = 10 , message = "name must be at least 10 characters ")

private String name ;

@NotBlank(message = "Password is required")

@Size(min = 8, message = "Password must be at least 8 characters")

@Pattern(regexp = ".*[@!\$#].*", message = "Password must contain at least one special character (@ ! \$ #)")

private String password ;

@NotBlank(message = "Email is required")

@Email

private String email ;

@Pattern(regexp = "^05\\d{8}\$", message = "Phone number must be 10 digits and start with 05")

private String phoneNumber;

Order class :

Represents a purchase order made by a customer .

Variables :

- orderId
- DateOfOrder
- status
- shipping address
- payment method

Validation :

```
@NotBlank(message = "Order ID is required")
private String orderId;
```

```
@NotBlank(message = "Date of order is required")
private String dateOfOrder;
```

```
@NotBlank(message = "Status is required")
@Pattern(
    regexp = "pending|shipped|cancelled",
    message = "Status must be pending, shipped, or cancelled"
)
private String status;
@NotBlank(message = "Shipping address is required")
private String shippingAddress;
```

```
@NotBlank(message = "Payment method is required")
@Pattern(
    regexp = "CASH|CARD",
    message = "Payment method must be CASH or CARD"
)
private String paymentMethod;
```